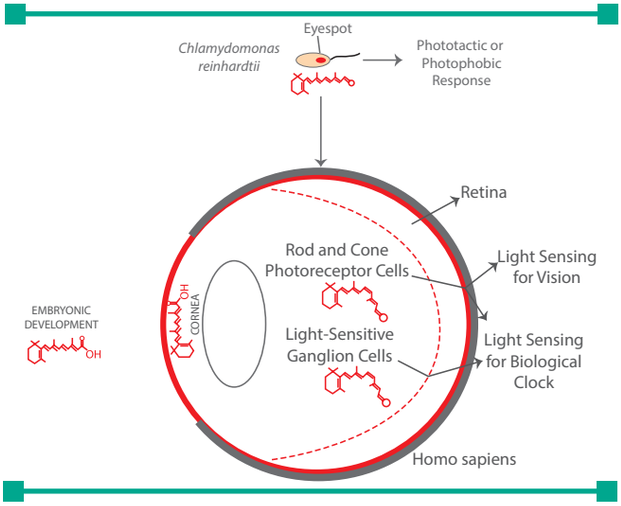
growth, normal function of the body and normal body processes.

5.3.2.1 Fat Soluble Vitamins

(a) **Vitamin A:** The vitamin A compounds include retinol, retinal and retinoic acid. Because it has a specific function in the retina of the eye and because it is an alcohol it was given the name retinol. Beta carotene is precursor of vitamin A and is found in large quantities in vegetables and fruits.

- **Functions:** Vitamin A performs the following functions:
  - Vitamin A is essential for vision in normal and dim light.
  - Formation and maintenance of healthy functioning epithelial tissue.
  - Glycoprotein and mucoprotein synthesis
  - Cancer prevention
  - Prevention of degeneration of myelin sheath
  - Normal bone formation and reproduction.



- **Food sources:** In the animal foods vitamin A is present in the form of retinol which are identified to be liver, cream, butter and egg yolk. Liver oils of fish like cod, halibut and shark are the richest sources of vitamin A. The main contributors of beta carotene are the yellow and green vegetable fruit sources of carotene- carrots, papaya, mango, sweet potatoes, spinach and broccoli.



- **Deficiency:** Decreased Vitamin A intake leads to vision problems
  - **Night blindness:** People suffering from night blindness cannot see objects in dim light
  - **Xerosis Conjunctiva:** The conjunctiva is dry, thickened, wrinkled and pigmented. This is due to the keratinization of the epithelial cells.
  - **Xerosis Cornea:** This manifests in the form of corneal dryness which gives the cornea a dull

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